

# Finite-depth Jacobian growth in GNNs

What random probes can estimate, what adjoints change, and how far forward-only methods reach

2026-09-29

## Executive summary

### The question

For a trained graph neural network, how should we estimate the largest depth-wise growth of a hidden-state perturbation, and what accuracy is actually attainable with forward-only random probes? This report studies a finite-depth quantity on controlled synthetic graphs, validates competing estimators against exact small-system references, and measures wall-clock cost.

### The quantity is finite-depth and unprojected

For a fixed graph  $G$ , input  $x$ , trained parameters  $\Phi$ , and hidden-state trajectory  $h_0 \mapsto h_K$ , define

$$B_K = D_{h_0}(F_{K-1,G} \circ \dots \circ F_{0,G})(h_0) = J_{K-1} \dots J_0, \quad \lambda_K^{\text{top}} = \frac{1}{K} \log \sigma_1(B_K).$$

The derivative includes the complete implemented layer: activation, normalization, residual paths, and state-dependent geometry. It excludes encoder and readout derivatives. No mean-centering or between-layer projection is inserted. Pierre’s projected cocycle is a separate observable, not an estimator of  $B_K$ .

This is a finite-depth sensitivity rate. It is not automatically an asymptotic Lyapunov exponent, an end-to-end prediction sensitivity, an oversquashing diagnostic, or a task-utility score.

### The main conclusions

1. **Random full-space probes are valid lower bounds but poor point estimators in this pilot.** The maximum over  $M$  independent JVP probes is monotone in  $M$  and converges almost surely for a fixed finite product, but it remained far from the exact rate after  $M = 2^{20}$  on the recent same-cell study: gaps were 0.26–0.35 for GCN/ChebNet and 0.085–0.097 for the sheaf baseline. The earlier broad calibration already found zero target coverage at  $M = 1024$  for every reported model/family cell.
2. **If VJPs are available, short operator-refined probes are the practical solution.** Power iteration on  $B_K^* B_K$  is an adjoint-enabled refinement of a probe, not a new finite-depth target. On the nine recent operators, eight of nine reached gap  $\leq 10^{-4}$  by step 4; six of nine reached  $\leq 10^{-12}$  by step 8. Early stopping and Lanczos reduced matrix-free wall-clock by 5–15 $\times$  relative to fixed three-start, 160-step power.
3. **Under a strict JVP-only restriction, graph-smoothed probes were the best tested variant.** Drawing full-space probes from  $q \propto (I + cL)^{-1}z$  improved the gap

at  $M = 2^{20}$  by about  $1.8\text{--}2.1\times$  relative to iid probes at similar wall-clock. A small- $M$  follow-up found that even ten smoothed probes gave pooled-best gaps of 0.17 (GCN), 0.21 (ChebNet), and 0.08 (sheaf); at  $M = 2^{10}$  these were about 0.09, 0.09, and 0.04. These are useful screening bounds, not tight estimates.

4. **The forward-only limitation is still a large gap.** The smoothed-probe curves retained the descriptive  $A - B \log_2 \log_2 M$  shape; extrapolations for a 0.01 gap were  $2^{233}$ – $2^{258}$  probes and are explicitly treated as out-of-range fits, not universal theorems. Forward-only  $B$ -power was also tested and stalled at eigenvalue/spectral-radius behavior on these nonnormal products rather than converging to  $\sigma_1(B_K)$ .
5. **Exact rates distinguish the trained baselines, but do not diagnose over-squashing alone.** Across 900 graph/signal draws per checkpoint, the mean exact top rates were GCN 0.804, ChebNet 1.024, Stable-ChebNet 0.281, sheaf 0.514, and heat diffusion approximately 0. Positive global amplification can coexist with contraction of most singular modes or with poor source-to-target communication.
6. **Stable-ChebNet is substantially less expansive in this controlled task, while retaining a smooth full-rank spectrum.** Its one-checkpoint histogram mean was 0.281, with family means 0.224/0.368/0.250 for homophilic/heterophilic/barbell draws; its complete-product spectra had no zero singular values in 58 graph-signal examples and a training median finite rate of -0.764. These are checkpoint- and hyperparameter-specific observations, not a claim that the paper’s architecture universally dominates.
7. **Pierre’s projection answers a different question.** The literal node-mean projected rate was lower than the full-space rate for every original architecture (GCN 0.344 vs 0.674, sheaf 0.427 vs 0.523, ChebNet 0.844 vs 1.112, matched-cell means), but it is the rate of a projected cocycle and must remain labeled separately.
8. **The expected empirical singular-value measure exposes the whole finite-depth spectrum.** Reverse-mode full-product SVDs on 290 graph-signal pairs (including every architecture’s 16 training examples and held-out validation/test distributions) show that the top-rate ordering persists, while the bulk is strongly contracting: training top-rate means are 0.777 (GCN), 0.987 (ChebNet), 0.136 (Stable-ChebNet), 0.505 (sheaf), and approximately 0 (heat), but the median finite log-rate over nonzero singular values is -11.98, -0.59, -0.76, -0.35, and -0.33 respectively. The report now shows both the expected CDF and its continuous PDF component, with zero-singular-value atoms annotated separately. This is a descriptive fixed-depth spectral measure, not a spectrum of an expected Jacobian or an asymptotic Lyapunov distribution.

The strongest practical recommendation is therefore conditional: use dense SVD for tiny calibration systems; use residual-checked  $B_K^* B_K$  power or Lanczos when adjoints are available; and use graph-smoothed full-space multiprobe only as an explicitly labeled JVP-only lower bound. No universal probe count is supported.

## Authoring and execution context

This revision was prepared in the **pi coding-agent harness**, session ID 01a0ceaa-cf63-77a8-8431-38ac76a7a7b5. Jacobian calculations, tests, plotting, and PDF builds were executed on Slurm compute nodes; file inspection and version-control operations were performed on the login node. The final source commit, lockfile hash, job IDs, manifests, and raw-result paths are recorded in RESEARCH\_LOG.md.

## 1. Estimand and estimator families

### 1.1 The hidden-state product

Let  $h_0 = e_G(x)$  be the encoder output and let

$$h_{\ell+1} = F_{\ell,G}(h_\ell), \quad \ell = 0, \dots, K-1.$$

The product is

$$B_K = J_{K-1} \cdots J_0, \quad J_\ell = D_{h_\ell} F_{\ell,G}(h_\ell),$$

and the target is

$$\lambda_K^{\text{top}}(G, x; \Phi) = K^{-1} \log \sigma_{\max}(B_K).$$

All norms are Euclidean/Frobenius norms after flattening the hidden tensor. Graph structure, parameters, evaluation-mode stochastic state, and the forward trajectory are frozen. For state-dependent sheaf layers, the derivative includes the path  $h \rightarrow R(h) \rightarrow L_{\mathcal{F}(h)}h$ ; detaching the geometry changes the model being measured.

The target does not include encoder/readout derivatives. If end-to-end input-output sensitivity is desired, it must be reported separately as a derivative of the complete prediction map. A finite feed-forward network also does not define an asymptotic Lyapunov exponent merely by being written as a sequence of layers; an asymptotic claim needs an extendable cocycle and a specified limiting regime.

## 1.2 What each estimator returns

| Method                             | Output                                                      | Strength                                           | Limitation and use                                                                |
|------------------------------------|-------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------|
| Dense Jacobian plus SVD            | Exact $\lambda_K^{\text{top}}$                              | Trusted tiny-system reference and full spectrum    | $D^2$ storage and roughly $D$ directional derivatives; not scalable               |
| Full-space multiprobe JVP          | $\max_{m \leq M} K^{-1} \log \ B_K q_m\ $                   | JVP-only, parallel, monotone lower bound           | No finite universal error certificate; can be very loose                          |
| Power on $B_K^* B_K$               | Approximation to $\lambda_K^{\text{top}}$                   | Few operator products; residual directly checkable | Requires VJPs and a correct adjoint through every layer                           |
| Matrix-free Lanczos on $B_K^* B_K$ | Ritz approximation to the same top singular value           | Often fewer iterations; residual available         | Requires VJPs and reorthogonalization                                             |
| Graph-smoothed multiprobe          | JVP-only lower bound with graph-informed probe distribution | Best forward-only variant tested here              | Distribution is a heuristic; c-scan is limited to this pilot                      |
| Forward-only $B$ -power            | Growth after repeatedly applying $B_K$                      | Cheap JVP-only refinement attempt                  | Targets eigenvalue behavior for nonnormal products, not generally $\sigma_1(B_K)$ |

## 1.3 The two important distinctions

First, a finite- $M$  multiprobe estimate is a lower bound:

$$\hat{\lambda}_{K,M} = \max_{m \leq M} \frac{1}{K} \log \|B_K q_m\| \leq \lambda_K^{\text{top}}.$$

For a nested iid stream it is nondecreasing in  $M$  and converges almost surely for each fixed finite product. It is not unbiased and does not certify a small gap at finite  $M$ .

Second, a power step is not a forward-only probe step. Applying  $B_K^* B_K$  requires a forward JVP sweep and a reverse VJP sweep. That extra adjoint is exactly what lets the algorithm seek the dominant right-singular direction rather than merely wait for a random probe to land near it.

## 2. Study design and evidence tiers

### 2.1 Architectures

The controlled baselines are intentionally small diagnostic models:

- **GCN:** depth 3,  $h_{\ell+1} = \text{ReLU}(\tilde{A}h_{\ell}W_{\ell})$ , with self-looped symmetric normalized adjacency.
- **ChebNet:** depth 3, order-3 Chebyshev filtering of the self-loop-free symmetric normalized Laplacian, with the configured degree-3 polynomial contrast.
- **Feature-conditioned diagonal-sheaf baseline:** depth 10, endpoint-dependent diagonal restriction maps, assembled sheaf-Laplacian action, residual update and ReLU. It is inspired by Neural Sheaf Diffusion but is not claimed to reproduce every detail of that paper; its implemented normalization is one-sided row normalization.
- **Stable-ChebNet (Hariri et al.):** depth 6, order-4 Chebyshev recurrence,  $\epsilon = 0.7$ ,  $\gamma = 0.01$ , antisymmetric channel weights with fixed dissipation, forward-Euler residual update, and tanh activation. This is a paper-inspired model-zoo implementation of arXiv:2506.07624v2, not a reproduction of the paper’s full long-range benchmark.

All models use four-dimensional inputs and hidden width 12 in the retained checkpoints. Exact equations and implementation choices are in `notes/experiment-baselines.md`.

### 2.2 Data and training

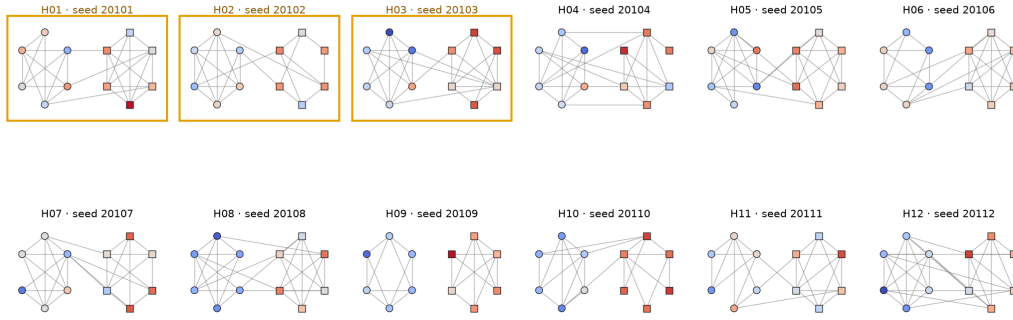
The supervised pilot uses deterministic synthetic node-classification data. Each node receives  $(2y_i - 1)e_1 + \epsilon_i$  with  $\epsilon_i \sim \mathcal{N}(0, I_4)$ . The three graph conditions are:

- a 12-node homophilic contextual SBM,  $p_{\text{same}} = 0.75$ ,  $p_{\text{other}} = 0.10$ ;
- a 12-node heterophilic SBM with those probabilities swapped;
- a 15-node barbell with two six-node cliques and a three-edge bridge.

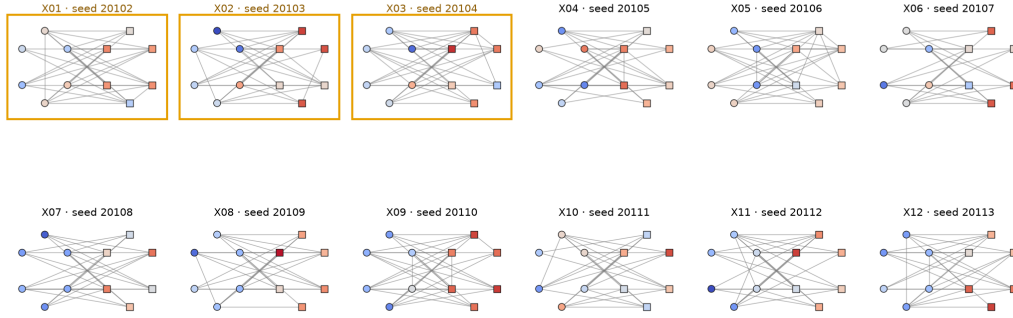
The split is 16 training graphs, 6 validation graphs, and 36 test graphs: 12 test graphs in each condition. Training uses 120 Adam epochs, weight decay  $10^{-4}$ , validation selection over widths  $\{8, 12\}$  and learning rates  $\{0.003, 0.01\}$ , and two model seeds. The held-out homophilic accuracy means were GCN 0.865, ChebNet 0.861, and sheaf 0.878; heterophilic accuracy was 0.312, 0.524, and 0.469 respectively. Stable-ChebNet used the paper-inspired fixed settings above, hidden width 12 and learning rate 0.003; its two seed-11/29 test accuracies were homophilic 0.896/0.910, heterophilic 0.514/0.493, and barbell 0.967/0.939, with seed 11 selected for downstream histograms and spectra. These figures establish that the checkpoints are trained task models, not a claim of broad architecture superiority.

All 36 held-out test graph/signal examples

#### Homophilic SBM



#### Heterophilic SBM



#### Barbell

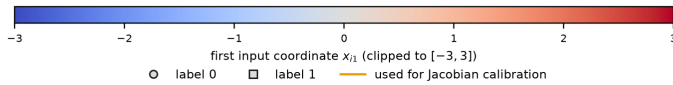
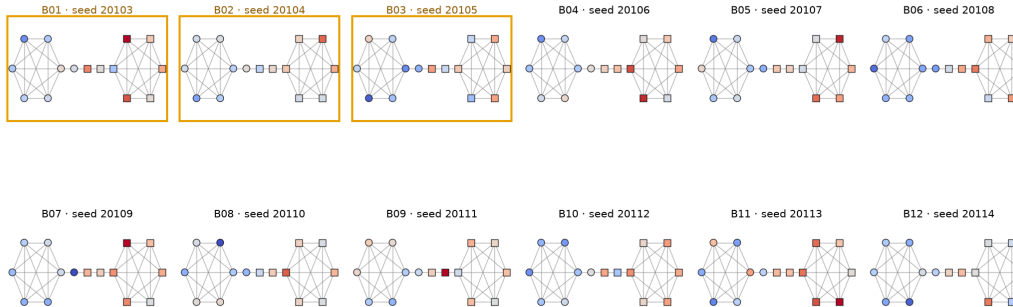


Figure 1. The 36 held-out graph/signal examples. Gold frames mark the nine unique calibration examples used for the main small-cell Jacobian study.

## 2.3 Evidence tiers and sample counts

The report combines several experiments; they should not be pooled as if they had the same scope.

| Evidence tier             | Operators / draws                                                                                                        | What it establishes                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Broad calibration         | 54 trained cells: 18 per architecture, from two model seeds, three families, and three held-out trajectories per family  | Initial probe, dense-reference, and power calibration                  |
| Recent probe sweep        | 9 operators: one seed-90000 example per family and one checkpoint per architecture; two independent streams per operator | $M = 2^{10} \dots 2^{20}$ convergence and same-cell method comparisons |
| Recent forward-only study | The same nine operators; iid and graph-smoothed streams, plus small- $M$ follow-up                                       | Wall-clock Pareto and tested JVP-only alternatives                     |
| Histograms                | 900 $(G, x)$ draws per one trained checkpoint/model                                                                      | Descriptive rate distributions and leading singular-mode distributions |
| Pierre comparison         | 54 matched projected cells                                                                                               | Separate projected-cocycle observable                                  |

The recent operator studies are deliberately narrower than the broad calibration. Their value is controlled comparison, not architecture-wide generalization. Stable-ChebNet received the same broad calibration, extended probe, timing, power-speedup, histogram, expected-spectrum, forward-only Pareto, c-scan, and adaptive follow-up evaluations; the Pierre comparison remains a separate projected-cocycle study of the original three baselines.

## 2.4 Validation and reproducibility

The implementation has tests for explicit-vs-JVP products, zero products, unit-probe enforcement, a GCN constant-mode/no-projection check, power-vs-dense linear products, nonlinear GCN/sheaf/Stable-ChebNet JVP products against independent reverse-mode dense autodiff, and Stable-ChebNet’s dissipative parameterization. The clean Stable-ChebNet validation at commit 0f78dd3 passed 26 tests; follow-up runner validation also passed on Slurm 22905. Float64 is used for calibration and comparison. The full lockfile hash, environment, seeds, checkpoints, graph conditions, job IDs, raw paths and artifact checksums are in each tracked manifest and RESEARCH\_LOG.md.

## 3. What random probes actually achieve

### 3.1 The broad calibration: $M \leq 1024$ is not a point-estimation budget

The initial nested-stream calibration used the full tangent space, no projection, independent streams, and exact dense-SVD references on tiny cells. At  $M = 1024$ , target coverage at gap  $\leq 0.01$  was 0/72 for every reported model/family cell. Pooled median gaps were approximately 0.395 for GCN and 0.109 for sheaf; the independent ChebNet replication gave medians 0.347, 0.346, and 0.376 for homophilic, heterophilic, and barbell families.

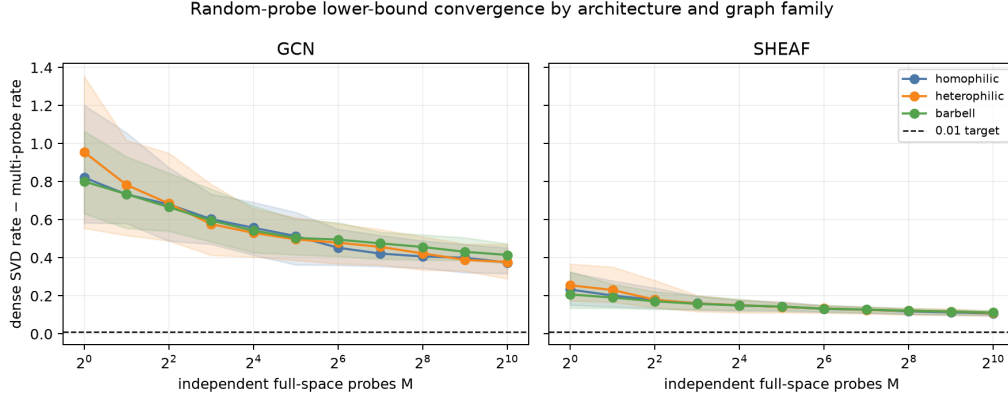


Figure 2. The initial  $M \leq 1024$  calibration. The maximum is a valid lower bound, but it is not close enough to the exact rate to be used as an unqualified point estimate in these cells.

### 3.2 The extended sweep: one million probes still leaves a visible gap

The extended run streamed iid full-space probes in chunks of 4096 to  $M = 2^{20} = 1,048,576$ . It completed all 18 streams without cutoff. One stream took roughly 9-11 seconds for GCN, 15-19 seconds for ChebNet, and 263-329 seconds for the depth-10 sheaf. Median gaps decreased only modestly from  $M = 2^{10}$  to  $2^{20}$ :

| Model          | $M = 2^{10}$ | $M = 2^{16}$ | $M = 2^{20}$ | Relative reduction |
|----------------|--------------|--------------|--------------|--------------------|
| GCN            | 0.390        | 0.355        | 0.318        | ~18% per 1024×     |
| ChebNet        | 0.380        | 0.344        | 0.308        | ~19% per 1024×     |
| Stable-ChebNet | 0.198        | 0.169        | 0.160        | ~19% per 1024×     |
| Sheaf          | 0.117        | 0.105        | 0.090        | ~23% per 1024×     |

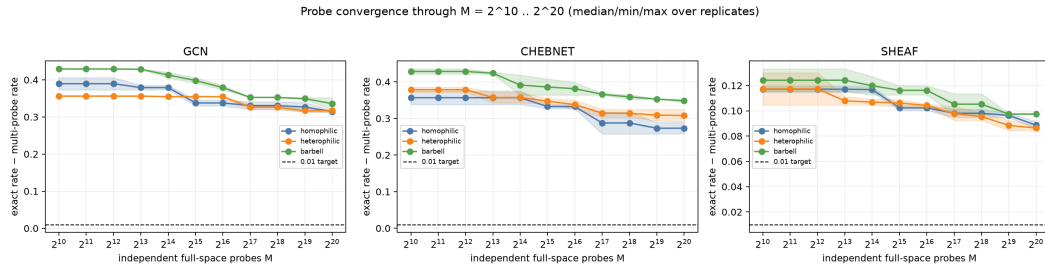


Figure 3. The extended iid-probe sweep for the original three baselines. The monotone lower bound improves, but the target gap is not approached at a feasible probe count.

The Stable-ChebNet replication reached  $2^{20}$  in all six streams. Its median gap decreased from 0.198 at  $2^{10}$  to 0.160 at  $2^{20}$ , with zero streams meeting the 0.01 target; the full curve and timings are shown below.

Probe convergence ( $M = 2^{10} \dots 2^{20}$ )

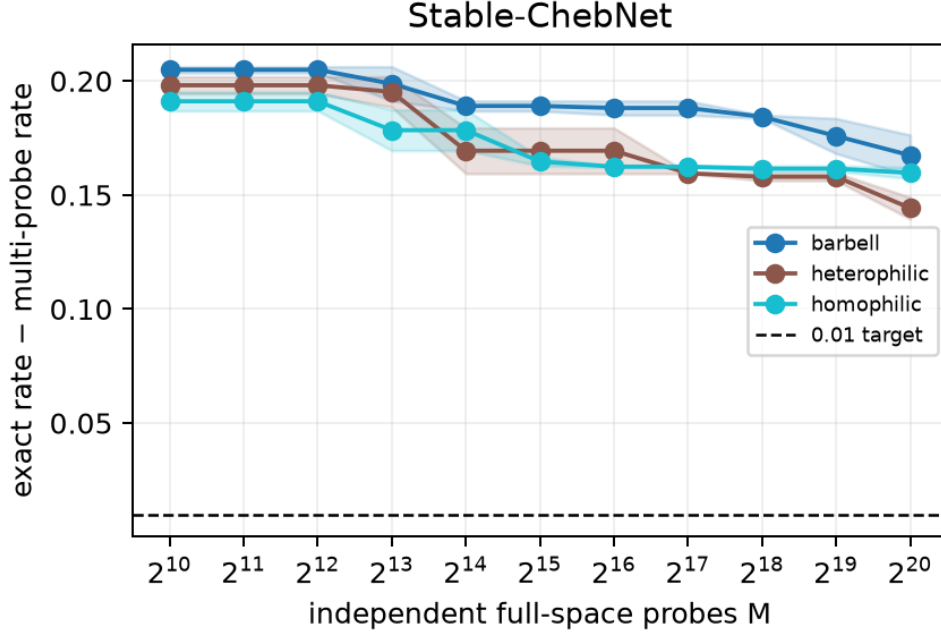


Figure 4. Stable-ChebNet iid-probe convergence on two independent streams per graph family. It is less difficult than the GCN/ChebNet baselines in this pilot, but a million probes still leaves a visible gap.

A fit of the form  $A - B \log_2 \log_2 M$  describes the measured range. Its extreme extrapolation should be treated as descriptive only: depending on the operator and distribution, the fitted  $M$  for a 0.01 gap lies around  $2^{245}$  or larger. This is evidence of an impractical overlap-concentration regime, not a universal theorem about all GNN Jacobians.

## 4. What adjoints change: operator-refined probes

### 4.1 Short $B_K^* B_K$ refinement

The same nine original operators and three Stable-ChebNet operators were used to compare random probes with an operator-refined probe. Starting from a random unit vector, each refinement applies

$$v \leftarrow \frac{B_K^* B_K v}{\|B_K^* B_K v\|}.$$

The rate is then evaluated from  $K^{-1} \log \|B_K v\|$ . This is ordinary power iteration on the symmetric positive semidefinite operator  $B_K^* B_K$ , expressed in the language of adaptive probes.

A probe-estimator variation that converges way faster: refine one probe instead of drawing  $10^6$

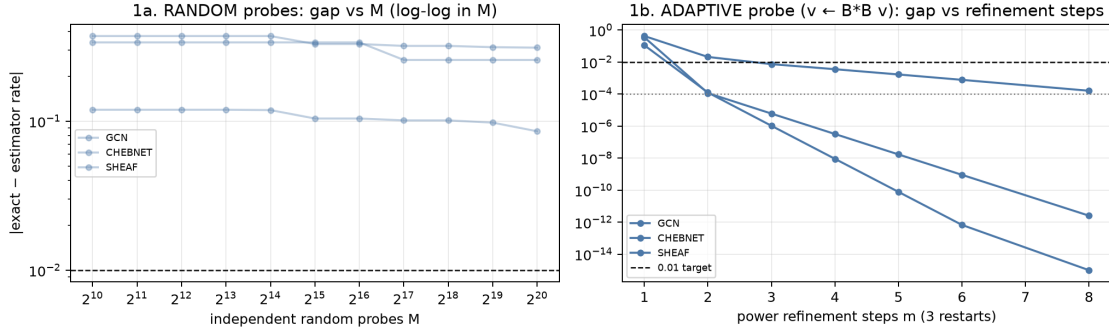


Figure 5. On identical operators, random probes improve slowly with  $M$ , while adjoint-enabled refinement reaches the reference in a handful of steps.

Across the nine original operators, 8/9 reached rate gap  $\leq 10^{-4}$  by step 4; 6/9 reached  $\leq 10^{-12}$  by step 8. The slow outlier was a ChebNet/homophilic operator whose gap remained about  $1.6 \times 10^{-4}$  at step 8. Stable-ChebNet reached  $10^{-4}$  in 3-5 steps on its three cells and  $10^{-12}$  in two of three by step 8. Thus the safe statement is “most cells reached  $10^{-4}$  in a few steps,” not that every cell did.

| Model          | Random gap at $2^{20}$ | Adaptive gap at step 2                         | Adaptive gap at step 3                         | Adaptive gap at step 6                          |
|----------------|------------------------|------------------------------------------------|------------------------------------------------|-------------------------------------------------|
| GCN            | 0.31-0.35              | $10^{-5}$ - $1.3 \times 10^{-3}$               | $6 \times 10^{-6}$ -<br>$3.1 \times 10^{-5}$   | about $10^{-12}$                                |
| ChebNet        | 0.26-0.34              | $1.3 \times 10^{-2}$ -<br>$2.1 \times 10^{-2}$ | $6 \times 10^{-6}$ -<br>$7.4 \times 10^{-3}$   | $1.7 \times 10^{-3}$ -<br>$10^{-11}$            |
| Sheaf          | 0.085-0.097            | $2.4 \times 10^{-5}$ -<br>$1.8 \times 10^{-4}$ | $5 \times 10^{-8}$ - $10^{-6}$                 | about $10^{-14}$                                |
| Stable-ChebNet | 0.149-0.176            | 0.0021-0.0048                                  | $2.7 \times 10^{-5}$ -<br>$9.1 \times 10^{-4}$ | $5.2 \times 10^{-11}$ -<br>$1.3 \times 10^{-5}$ |

Each refinement costs one forward JVP sweep and one reverse VJP sweep. This is why the convergence improvement is so large: the method actively aligns with the right-singular direction rather than waiting for a random draw. On Stable-ChebNet, the adaptive replication reached gap below  $10^{-4}$  in 3 steps for two families and 5 for the homophilic cell; it reached below  $10^{-12}$  by step 8 in two of three families. Its random-probe fit still extrapolated to an infeasible probe count for a 0.01 gap.

## 4.2 Matrix-free speedups and residual gates

For the same nine original operators plus three Stable-ChebNet operators, early-stopping power and Lanczos were compared with fixed three-start, 160-step power. Median seconds per unique operator, with three timed repeats, were:

| Model          | Fixed power | Early power | Lanczos | Early speedup | Lanczos speedup |
|----------------|-------------|-------------|---------|---------------|-----------------|
| GCN            | 1.26        | 0.086       | 0.12    | $14.7 \times$ | $10.4 \times$   |
| ChebNet        | 2.64        | 0.36        | 0.49    | $7.3 \times$  | $5.4 \times$    |
| Sheaf          | 30.7        | 2.06        | 5.9     | $14.9 \times$ | $5.2 \times$    |
| Stable-ChebNet | 9.20        | 0.612       | 2.79    | $15.0 \times$ | $3.3 \times$    |

The benchmark has three unique operators per architecture and three timed repeats per operator. All timed early-power and Lanczos calls passed their residual gate and matched the exact small-system reference to numerical precision. At these tiny tangent dimensions, dense SVD is still faster; the matrix-free result matters for the dimension regime where dense storage is impossible.

A fine-grained stopping experiment also compared residual thresholds  $10^{-7}$  and  $10^{-2}$ . With checks every 10 sweeps, the thresholds often stopped at the same checkpoint. Checking every sweep let the looser threshold stop earlier: median times were 0.045, 0.136, and 1.09 seconds for GCN, ChebNet, and sheaf, with maximum observed rate gaps  $3.1 \times 10^{-5}$ ,  $2.5 \times 10^{-4}$ , and  $2.4 \times 10^{-5}$  in that timing sample. This is a calibrated speed/accuracy option, not a universal residual-to-rate error theorem.

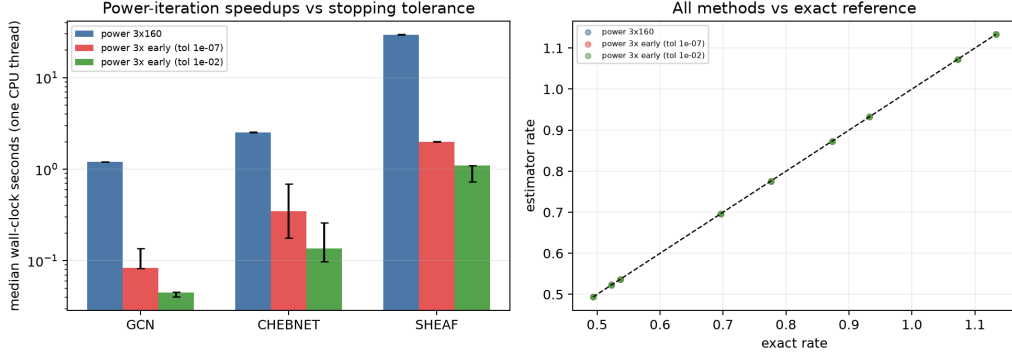


Figure 6. The fixed, early-stopped, loose-threshold, and Lanczos timing comparison for the original three baselines.

Stable-ChebNet showed the same qualitative ranking: fixed power took 9.20 s per timed cell, early power with a  $10^{-7}$  residual gate took 0.612 s ( $15.0\times$  faster), and Lanczos took 2.79 s ( $3.3\times$  faster). The Stable-specific timing distribution is included here:

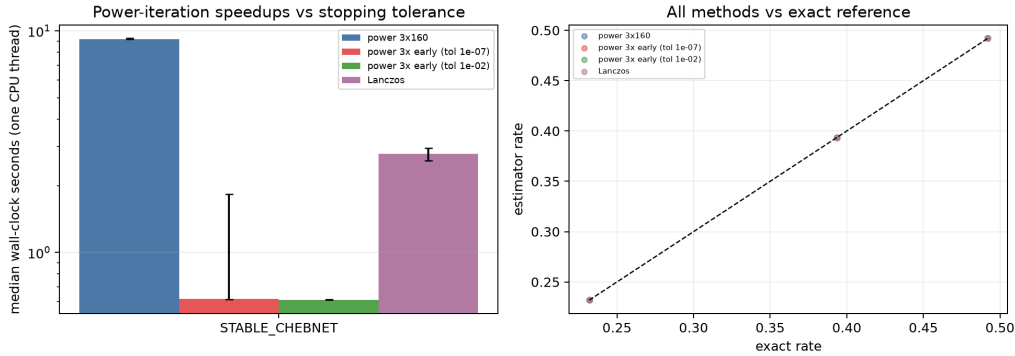


Figure 7. Stable-ChebNet fixed-power, early-stopped power, loose-threshold power, and Lanczos timings. All nine cells matched the dense reference to numerical precision; the looser  $10^{-2}$  threshold stopped at the same ten-sweep checkpoint in this sample.

## 5. The best tested JVP-only method

### 5.1 Graph-smoothed full-space probes

The strict forward-only setting permits JVP applications of  $B_K$  but no adjoint. In that setting, forward  $B$ -power is not a valid replacement for singular-value iteration on general nonnormal products: its direction is eigenvalue-driven and empirically stalled below the top singular rate by gaps of 0.11–0.57 after 40 steps on these cells.

The remaining tested lever is the probe distribution. We drew full-space probes using

$$q = \frac{(I + cL)^{-1}z}{\|(I + cL)^{-1}z\|},$$

where  $L$  is the normalized graph Laplacian and  $z$  is Gaussian. For finite  $c$ , this remains a full-support distribution and therefore preserves the same asymptotic target in principle; it is a graph-informed importance distribution, not a projection.

On the same nine recent operators,  $c = 1$  smoothed probes dominated iid probes at every recorded marker. At  $M = 2^{20}$ :

| Model          | iid gap | Smoothed gap | Median wall-clock at $2^{20}$ |
|----------------|---------|--------------|-------------------------------|
| GCN            | 0.313   | <b>0.166</b> | 8.7 / 9.8 s                   |
| ChebNet        | 0.292   | <b>0.154</b> | 15.0 / 16.2 s                 |
| Sheaf          | 0.086   | <b>0.045</b> | 280.5 / 280.9 s               |
| Stable-ChebNet | 0.157   | <b>0.109</b> | 37.5 / 40.0 s                 |

The  $c$ -scan was performed on one homophilic operator per architecture, at  $M = 2^{14}$ , and is therefore a local sensitivity check rather than a universal tuning rule. Stable-ChebNet was evaluated on the same cell:

| $c$     | GCN          | ChebNet      | Sheaf        | Stable-ChebNet |
|---------|--------------|--------------|--------------|----------------|
| 0 (iid) | 0.373        | 0.338        | 0.119        | 0.169          |
| 0.5     | 0.262        | 0.245        | 0.090        | 0.143          |
| 1       | 0.205        | 0.197        | 0.076        | 0.128          |
| 4       | 0.109        | 0.114        | 0.051        | 0.104          |
| 16      | <b>0.091</b> | <b>0.077</b> | <b>0.039</b> | <b>0.080</b>   |

Best forward-only estimator wall-clock Pareto (forward probes only; rightmost point =  $M = 2^{20}$ )

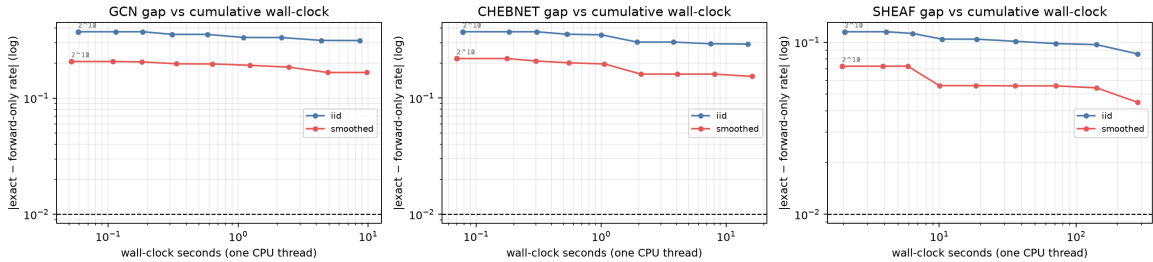


Figure 8. Gap versus cumulative wall-clock for iid and graph-smoothed forward-only probes. Probe work is batched; the total wall-clock still grows with  $M$ , especially for the deep sheaf.

The marginal batched probe costs were approximately 8 microseconds for GCN, 14 microseconds for ChebNet, and 0.25 milliseconds for sheaf in this CPU benchmark. The Stable-ChebNet  $c$ -scan improved the homophilic  $M = 2^{14}$  gap monotonically from 0.169 at  $c = 0$  to 0.080 at  $c = 16$ . Its full  $2^{20}$  replication reached 37.5 s (iid) or 40.0 s (smoothed), with gaps 0.157 and 0.109 respectively; its forward-only  $B$ -power plateau gaps were 0.16–0.25. The important limitation is that the gap decreases too slowly for this cost to buy a 0.01 estimate.

## 5.2 Very small probe counts

A follow-up measured  $M \in \{10, 32, 128, 512, 1024, 4096, 16384\}$  with five independent streams on one homophilic operator per architecture. The table reports the median gap

across streams and, in parentheses, the best pooled gap among those five streams; the pooled value is optimistic and should not be confused with a typical run.

| Model,<br>smoothed<br>$c = 16$ | $M = 10$    | $M = 2^{10}$ | $M = 2^{12}$ | $M = 2^{14}$ |
|--------------------------------|-------------|--------------|--------------|--------------|
| GCN                            | 0.30 (0.17) | 0.15 (0.09)  | 0.12 (0.09)  | 0.09 (0.08)  |
| ChebNet                        | 0.28 (0.21) | 0.13 (0.09)  | 0.11 (0.09)  | 0.09 (0.08)  |
| Sheaf                          | 0.09 (0.08) | 0.045 (0.04) | 0.042 (0.03) | 0.039 (0.03) |
| Stable-<br>ChebNet             | 0.17 (0.14) | 0.09 (0.08)  | 0.08 (0.08)  | 0.08 (0.08)  |

For comparison, the GCN iid median gaps were 0.61, 0.41, and 0.37 at  $M = 10$ ,  $2^{10}$ , and  $2^{14}$  respectively. A frequency-adaptive reweighting of probes toward nodes used by the best probes was also tested. It spent probes learning the reweighting and was worse than fixed smoothing at every tested count; for example, GCN at  $M = 10$  was 0.37 adapted versus 0.17 for the pooled-best smoothed stream. Stable-ChebNet showed the same negative result: at  $M = 10$ , the pooled gaps were 0.20 for adapted versus 0.14 for fixed smoothed probes, and at  $M = 2^{14}$  they were 0.078 versus 0.074. Fixed graph smoothing is therefore the best JVP-only distribution tested here, not a proof of a universal optimum.

### 5.3 The practical JVP-only conclusion

The forward-only result is deliberately modest:

- graph-smoothed probes are a real, reproducible improvement over iid probes on these synthetic operators;
- ten probes can give a useful coarse lower bound, especially for the sheaf baseline, but not a tight top-rate estimate;
- at  $M = 2^{10}$ , the best tested pooled gaps remain about 0.04–0.09 for the displayed operators;
- the fitted double-log behavior means that a 0.01 target is not feasible in this study by forward-only probing alone.

This should be reported as a **lower-bound screening method**. If the scientific question requires a tight top singular rate, the correct escape is an adjoint-enabled method, not more confidence in a large finite- $M$  maximum.

## 6. What the exact rates reveal about these models

### 6.1 Rate distributions

The histogram diagnostic loads checkpoints and never trains. It draws the same 900 graph/signal pairs per trained checkpoint: 300 per family. The exact reverse-mode dense Jacobian/SVD rates have means:

| System             | Mean               | Median             | Standard deviation          | Homophilic         | Heterophilic       | Barbell            |
|--------------------|--------------------|--------------------|-----------------------------|--------------------|--------------------|--------------------|
| GCN                | 0.804              | 0.793              | 0.068                       | 0.792              | 0.825              | 0.796              |
| ChebNet            | 1.024              | 1.028              | 0.076                       | 0.996              | 1.091              | 0.984              |
| Stable-<br>ChebNet | 0.281              | 0.285              | 0.124                       | 0.224              | 0.368              | 0.250              |
| Sheaf              | 0.514              | 0.517              | 0.026                       | 0.506              | 0.528              | 0.507              |
| Heat<br>diffusion  | approximately<br>0 | approximately<br>0 | approximately<br>$10^{-17}$ | approximately<br>0 | approximately<br>0 | approximately<br>0 |

Exact top (1st) eigenmode rates over random  $(G, x)$  pairs (same 300 draws/family)

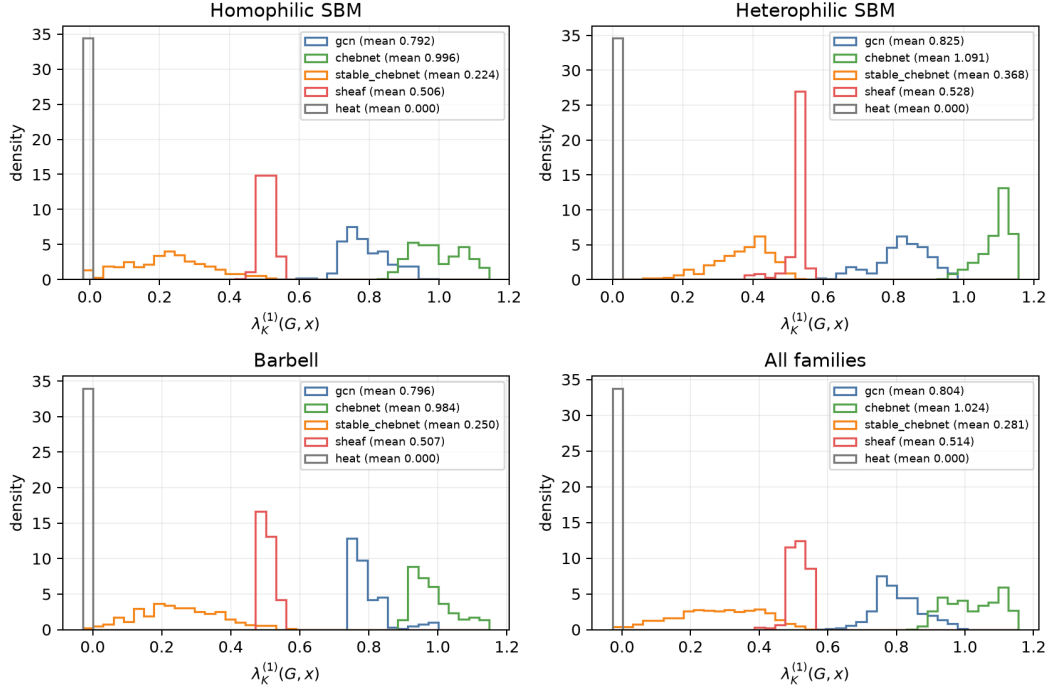


Figure 9. Exact finite-depth top-rate distributions for one trained checkpoint per GNN, Stable-ChebNet, and the heat control. These are descriptive draws, not population-level architecture claims.

ChebNet has the largest mean top rate, followed by GCN and the sheaf baseline. The distributions are narrow on these synthetic draws, but one checkpoint per architecture is not enough to establish a general architecture ordering. Heat diffusion provides a non-expanding control: with the symmetric normalized Laplacian, the preserved mode is a normalized harmonic mode, not necessarily the raw constant vector; the remaining modes contract.

## 6.2 Leading singular modes

The exact diagnostic also computed the leading 25 singular values per draw. The following table uses modes 1, 13, and 25 for GNNs and modes 1, 2, and 3 for heat:

| System         | Mode 1          | Mode 13 / 2 | Mode 25 / 3 | Interpretation                                  |
|----------------|-----------------|-------------|-------------|-------------------------------------------------|
| GCN            | 0.804           | -0.733      | -1.473      | Growth concentrated in a few directions         |
| ChebNet        | 1.024           | 0.389       | 0.248       | Positive mean persists through reported mode 25 |
| Stable-ChebNet | 0.281           | -0.028      | -0.156      | Smooth decay after the leading mode             |
| Sheaf          | 0.514           | 0.125       | 0.040       | Gradual spectral decay                          |
| Heat           | approximately 0 | -0.072      | -0.155      | Harmonic mode preserved; other modes contract   |

Eigenmode rate histograms per system (GNNs: modes 1/13/25; heat: modes 1/2/3)

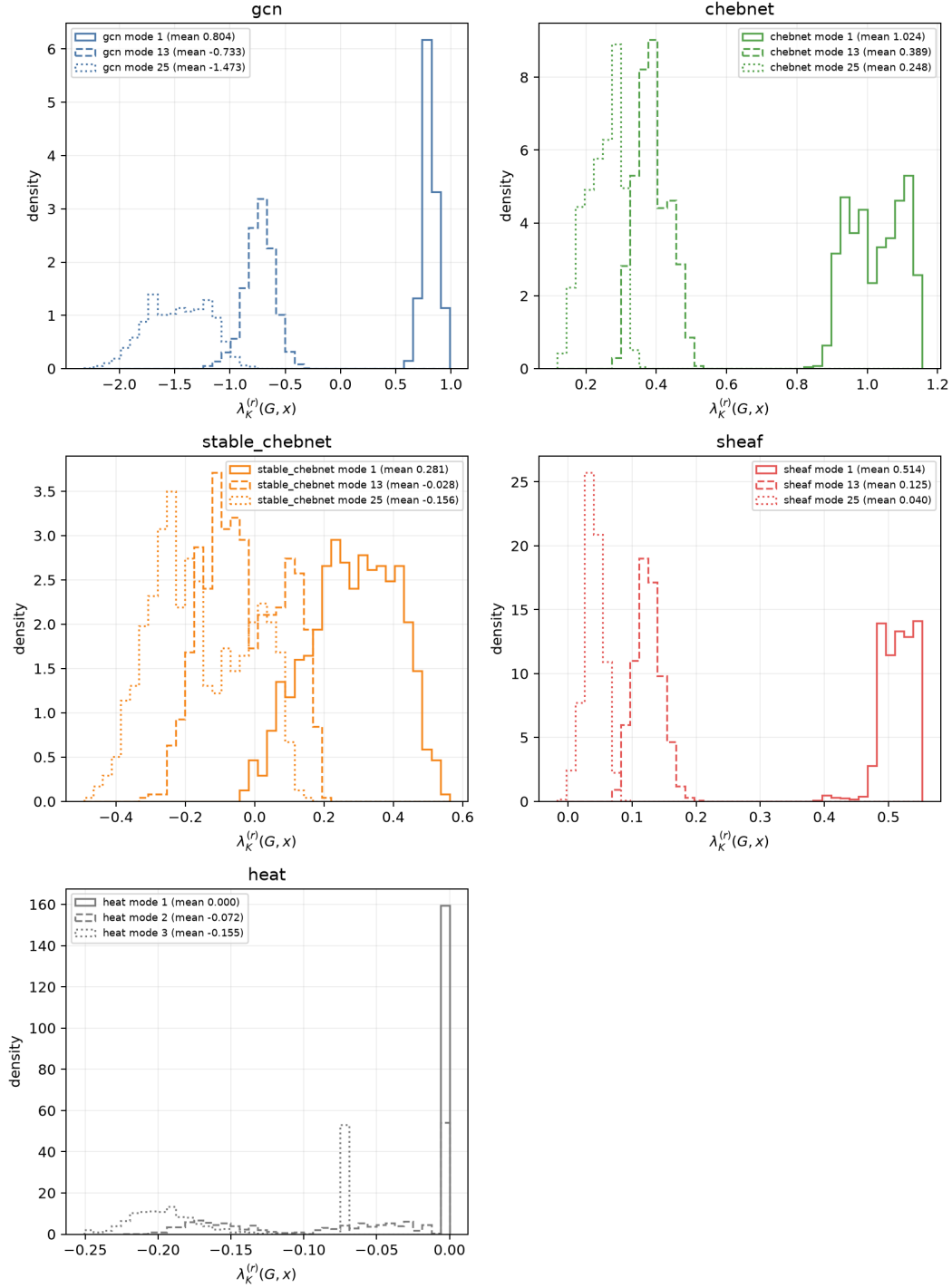


Figure 10. These are singular-mode rates, not eigenvalue rates. “Positive through mode 25” refers to the reported mean, not every individual draw.

A positive global top rate is compatible with contraction of most directions. It also says nothing by itself about individual source-to-target Jacobian blocks, delocalization, oversquashing, or classification performance. Those require separate observables.

### 6.3 Expected empirical singular-value measures

For each sampled pair, let  $s_1 \geq \dots \geq s_d \geq 0$  be the singular values of the **complete** hidden-state product  $B_K$ . We define the empirical measure  $\mu_{G,x} = d^{-1} \sum_i \delta_{s_i}$  and average these measures over graph-signal pairs. The primary plotted representations are the expected CDF of the per-depth log singular values,  $\bar{F}(t) = \mathbb{E}[d^{-1} \#\{i : K^{-1} \log s_i \leq t\}]$ , and the corresponding continuous PDF over finite rates. This gives each pair equal weight, retains zero modes, and is distinct from both the spectrum of  $\mathbb{E}[B_K]$  and the expected top singular value.

The new run used the explicit reverse-mode dense Jacobian of the complete composition followed by a full SVD, with no projections. It evaluated 16 training graphs, 6 validation graphs and 36 held-out test graphs for each GNN checkpoint, plus the same graph distributions for single-channel heat diffusion. The independent JVP-product check on the first cell had maximum absolute discrepancy  $2.2 \times 10^{-16}$ . The heat Jacobian is graph-only, so its signal draw is immaterial.

| System         | Training top-rate mean | Test top-rate means (hom./het./barbell) | Training zero-singular-value mass | Training median finite log-rate |
|----------------|------------------------|-----------------------------------------|-----------------------------------|---------------------------------|
| GCN            | 0.777                  | 0.768 / 0.782 / 0.805                   | 2.43%                             | -11.98                          |
| ChebNet        | 0.987                  | 1.014 / 1.073 / 0.987                   | 0.61%                             | -0.59                           |
| Stable-ChebNet | 0.136                  | 0.258 / 0.333 / 0.310                   | 0%                                | -0.76                           |
| Sheaf          | 0.505                  | 0.502 / 0.514 / 0.512                   | 0.56%                             | -0.35                           |
| Heat diffusion | approximately 0        | approximately 0 / 0 / 0                 | 0%                                | -0.33                           |

The extreme negative GCN median reflects its rapidly collapsing tail and near-zero singular values; it should not be read as a mean sensitivity of the model. The CDFs make the more robust statement visible: all systems have a small leading expanding or preserved mode, but the bulk contracts, with the GCN tail most sharply separated from its top mode. ChebNet retains the broadest relatively positive spectrum among the learned models in this pilot. These are checkpoint- and distribution-specific summaries, not architecture-wide claims.

Expected empirical singular-value measures of the complete depth- $K$  Jacobian

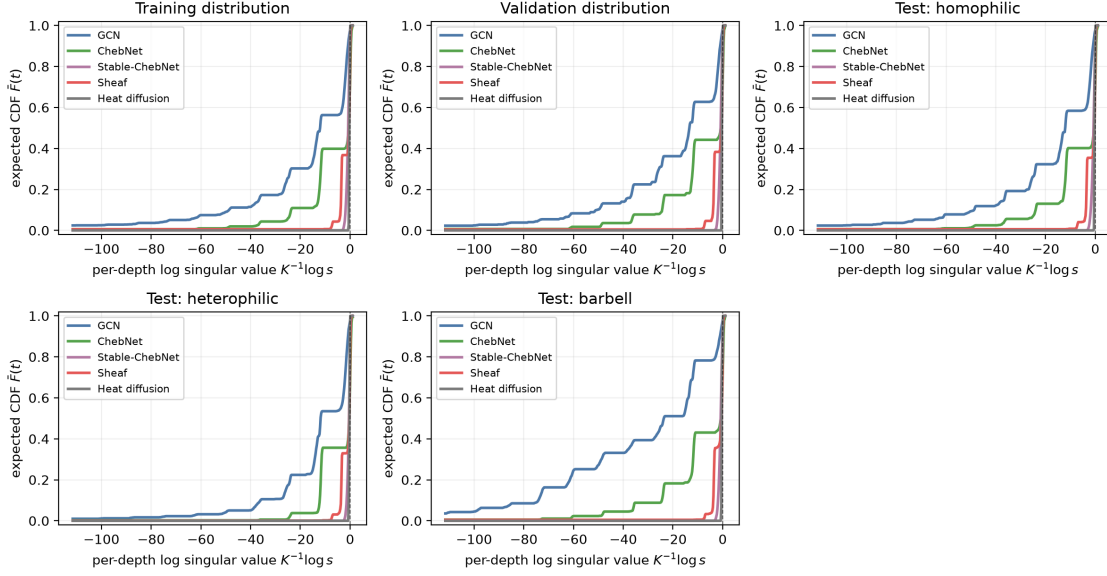
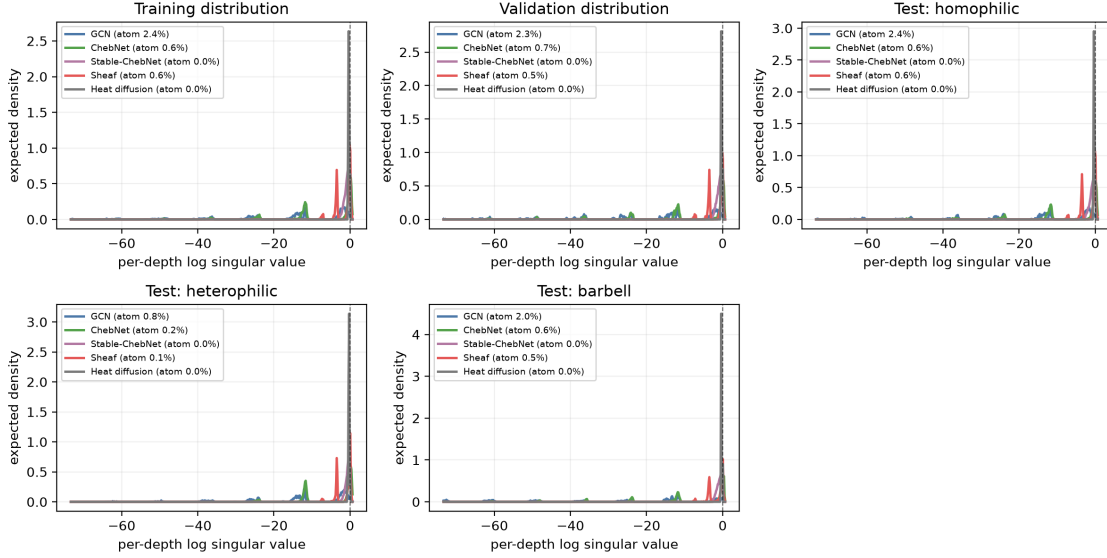


Figure 11. Expected CDFs of per-depth log singular values for the complete depth- $K$  Jacobian. Each curve averages per-example empirical CDFs; the training panel is included for every architecture, and the remaining panels show held-out distributions.

PDF component of expected empirical log-singular-value measures



Continuous density is shown over the central 98.5% finite-rate range; zero singular values are annotated as an atom at  $-\infty$ .

Figure 12. PDF component of the expected empirical per-depth log-singular-value measure. Curves are normalized by each pair's full tangent dimension and averaged across pairs; the continuous density is shown over the central finite-rate range, while zero singular values appear as annotated atoms at  $-\infty$ .

Mean spectral quantile profiles (all singular values, not only the top mode)

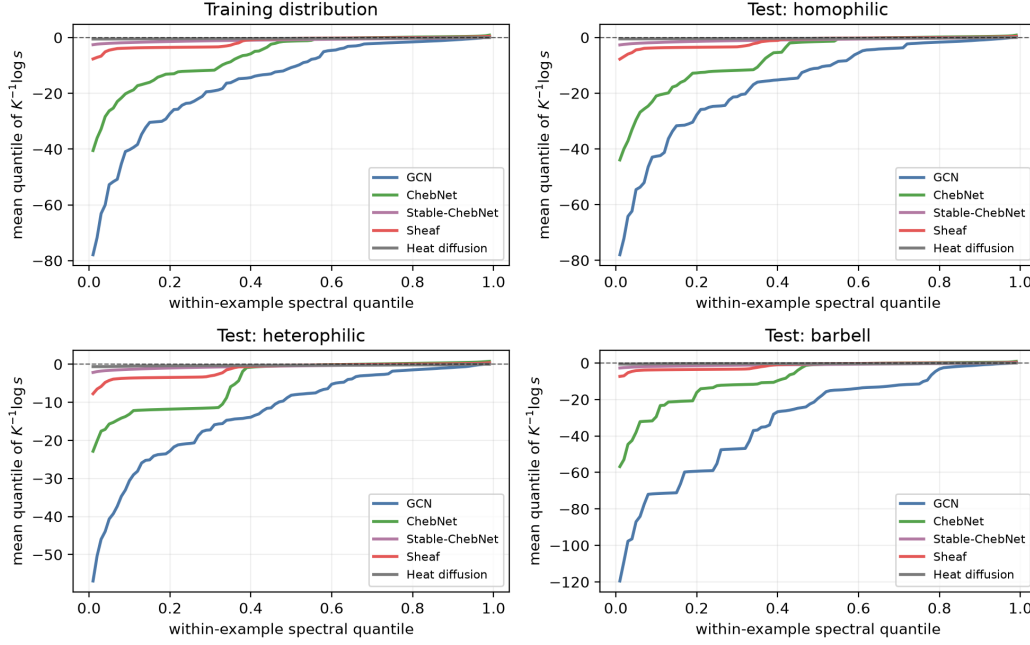


Figure 13. Mean within-example spectral quantile profiles. These curves complement the expected CDFs and PDFs and show how the bulk, not only the top mode, changes across systems and graph families.

## 7. Pierre’s projected observable

Pierre’s literal projected cocycle initializes probes in the node-mean-zero subspace and applies

$$B_K^P = PJ_{K-1}PJ_{K-2} \cdots PJ_0|_{\mathbf{1}_N^\perp}, \quad Pz = z - \frac{1}{N}\mathbf{1}\mathbf{1}^\top z$$

after every layer. This can be a meaningful transverse observable, but

$$B_K^P \neq B_K$$

in general. For normalized graph Laplacians the natural harmonic vector is degree-weighted, and for sheaves the harmonic space depends on the learned restriction maps; the literal node-mean prescription is a deliberate choice, not an invariant-subspace theorem.

The matched projected calibration used 54 cells per estimator family, 12 nested streams through  $M = 1024$ , and projected dense references. Mean exact rates were:

| Model          | Full-space $B_K$ | Projected $B_K^P$ | Difference |
|----------------|------------------|-------------------|------------|
| GCN            | 0.674            | 0.344             | -0.330     |
| Sheaf          | 0.523            | 0.427             | -0.096     |
| ChebNet        | 1.112            | 0.844             | -0.268     |
| Stable-ChebNet | 0.260            | 0.170             | -0.090     |

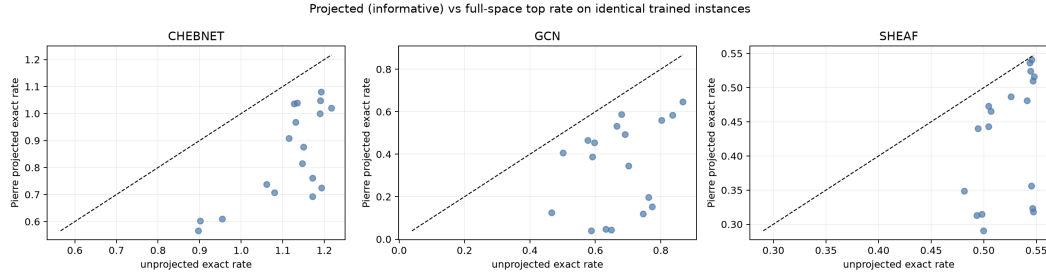


Figure 14. Projection lowers the measured top rate on these matched cells, but the projected quantity remains positive on average. The comparison is between two estimands, not two implementations of one estimator.

At  $M = 1024$ , projected probes also missed their exact target. Their finite- $M$  averages were GCN -0.038, sheaf 0.324, ChebNet 0.517, and Stable-ChebNet -0.022, versus their full-space values 0.280, 0.414, 0.759, and 0.260 respectively. These are lower bounds, not substitutes for the projected dense reference; the Stable-ChebNet projected calibration is tracked separately under 2026-09-29-pierre-estimator-calibration-stable-chebnet.

## 8. Recommended protocol

| Situation                             | Recommended estimator                                                                     | Acceptance rule and reporting                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Tiny calibration system               | Dense reverse-mode Jacobian/SVD, independently checked against JVP product                | Report $D$ , product discrepancy, singular spectrum, state boundary and precision                                                         |
| Tight finite-depth top rate with VJPs | Multi-start power on $B_K^* B_K$ with residual checks; use Lanczos if convergence is slow | Report each start, residual, iterations, JVP/VJP counts, accepted estimate and exact-SVD comparison where available                       |
| JVP-only screening                    | Batched full-space graph-smoothed multiprobe                                              | Report distribution, $c$ , $M$ , independent streams, wall-clock, and the nonnegative gap interpretation; never call it an exact exponent |
| Unknown or large system               | Calibrate on a smaller matched instance first                                             | Do not transfer a probe count or residual threshold without evidence                                                                      |
| Long stationary/extendable cocycle    | QR/Benettin or another asymptotic-spectrum method                                         | State the stationarity/extendability assumptions; do not identify a finite-network top singular rate with the asymptotic exponent         |
| Oversmoothing/oversquashing question  | Separate transverse, blockwise, or source-to-target observables                           | Global $\sigma_1(B_K)$ is insufficient                                                                                                    |

The operational defaults are therefore: dense reference when feasible; residual-checked matrix-free power/Lanczos when adjoints exist; graph-smoothed multiprobe as a JVP-only lower-bound screen. No projection  $P$  is inserted into the primary target.

## 9. Limitations and next experiments

- The graphs are small synthetic graphs selected for exact calibration. The recent nine-operator follow-ups and the  $c$ -scan are narrower than the broad calibration and do not establish a universal probe distribution or probe count.
- The sheaf model is a diagnostic implementation, not a claim to reproduce every Neural Sheaf Diffusion variant. The state-dependent- $\mu$  architecture remains underspecified in notes/state-dependent-mu-diffusion-GNN.md and is not silently treated as implemented.
- Stable-ChebNet is implemented from Hariri et al.’s published layer equations and checked against the paper’s qualitative stability mechanism, but this study uses one small synthetic task, one fixed hyperparameter setting, and one downstream checkpoint for most distributional evaluations; it is not a full reproduction of the paper’s long-range benchmark.
- Histograms use one trained checkpoint per model. Their model ordering and narrowness are descriptive for these draws, not architecture-level conclusions.
- Each recent  $(G, x)$  study varies graph and feature signal jointly. It does not estimate input-only variability on a fixed graph.

- The fitted  $A - B \log_2 \log_2 M$  curves are extrapolations beyond the measured range. They motivate caution but are not universal lower bounds.
- A positive global top rate does not prove long-range communication, delocalization, or absence of oversquashing. Future work should measure source-to-target Jacobian blocks, mode localization, and task-relevant subspaces separately.
- The next computational scale-up should keep the exact-SVD calibration suite, then test residual-gated power/Lanczos and forward-only graph-smoothed probes on larger graphs, real datasets, more checkpoints, and explicitly specified adaptive-geometry models.

## 10. Evidence map and reproducibility

Tracked evidence is under `evidence/<run-id>/`; raw outputs remain under `ignored results/<run-id>/` on the cluster. The most relevant artifacts are:

- 2026-09-24-probe-calibration-v2: broad  $M \leq 1024$  calibration and exact references, Slurm 22351.
- 2026-09-24-estimator-wall-clock: dense/probe/fixed-power timing, Slurm 22388.
- 2026-09-24-lyapunov-histogram-comparison: exact 900-draw model/heat histograms and leading singular modes, Slurm 22400/22406 and curation 22402/22407.
- 2026-09-24-pierre-estimator-calibration: matched projected/unprojected study, Slurm 22377/22380.
- 2026-09-26-probe-convergence-2p20: iid stream through  $2^{20}$ , Slurm 22803.
- 2026-09-26-power-speedup: early stopping and Lanczos, Slurm 22796/22803.
- 2026-09-27-probe-vs-adaptive: random versus adjoint-enabled refinement, Slurm 22828.
- 2026-09-27-forward-only-pareto: iid versus graph-smoothed JVP-only probes, Slurm 22846.
- 2026-09-27-forward-adapted: small- $M$  and frequency-adaptive negative result, Slurm 22855.
- 2026-09-28-expected-singular-measures: complete reverse-mode spectra on 232 training/validation/test graph-signal pairs for GCN, ChebNet, sheaf and heat, Slurm 22886.
- 2026-09-29-stable-chebnet-calibration: broad Stable-ChebNet calibration on two seeds and three graph families, Slurm 22896.
- 2026-09-29-expected-singular-measures-stable-chebnet: complete Stable-ChebNet spectra alongside the prior systems, Slurm 22898.
- 2026-09-29-lyapunov-histogram-stable-chebnet and 2026-09-29-lyapunov-histogram-comparison-stable: 900-draw Stable-ChebNet histogram and five-system comparison, Slurm 22899/22904.
- 2026-09-29-probe-convergence-stable-chebnet: Stable-ChebNet iid probes through  $2^{20}$ , Slurm 22900.
- 2026-09-29-estimator-wall-clock-stable-chebnet and 2026-09-29-power-speedup-stable-chebnet: Stable-ChebNet estimator timings and matrix-free speedups, Slurm 22902/22901.
- 2026-09-29-smooth-c-sweep-stable-chebnet: Stable-ChebNet graph-smoothing c-scan at  $M = 2^{14}$ , Slurm 22920.
- 2026-09-29-pierre-estimator-calibration-stable-chebnet: Stable-ChebNet's separate projected-cocycle calibration, Slurm 22925.

All experimental conclusions in this report point to tracked figures/tables and manifests. The complete command lines, commits, lockfile hash, Python/PyTorch versions, seeds, checkpoint paths, reference tolerances, failure modes, and Slurm logs are recorded in `RESEARCH_LOG.md`.

## References

Local-source provenance is indexed in papers/README.md. The core references are:

- Kipf & Welling, Semi-Supervised Classification with Graph Convolutional Networks, arXiv:1609.02907.
- Defferrard, Bresson & Vandergheynst, Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering, arXiv:1606.09375.
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- Oono & Suzuki, Graph Neural Networks Exponentially Lose Expressive Power for Node Classification, arXiv:1905.10947.
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- Hariri et al., Return of ChebNet: Understanding and Improving an Overlooked GNN on Long Range Tasks, arXiv:2506.07624v2.
- Johansson, Strannegård, Engsner & Mostad, Exact Spectral Norm Regularization for Neural Networks, arXiv:2206.13581.
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- Benettin et al., Lyapunov Characteristic Exponents for Smooth Dynamical Systems, Meccanica 15, 1980.